

ComplexBrowser: A Tool for Identification and Quantification of Protein Complexes in Large-scale Proteomics Datasets

Authors

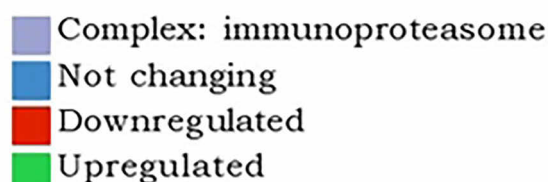
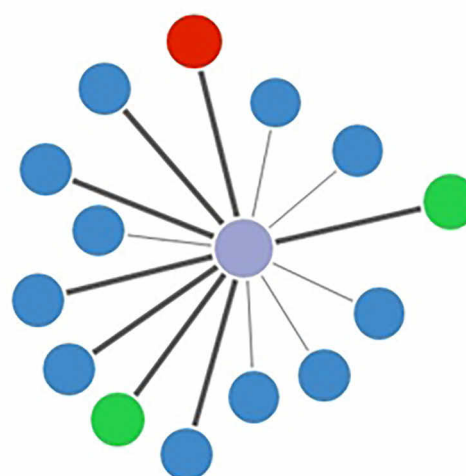
Wojciech Michalak, Vasileios Tsiamis, Veit Schwämmle, and Adelina Rogowska-Wrzesińska

Correspondence

adelinar@bmb.sdu.dk

Graphical Abstract

Quantitative analysis of protein complexes



In Brief

ComplexBrowser can identify protein complexes registered in manually curated databases that are present in datasets obtained from large scale quantitative proteomic experiments (label free or isobaric mass tag based). It provides, in the form of the CFC factor, a quantitative measure of the coordinated changes in complex components. This facilitates assessing the trends in the processes governed by the identified complexes providing a new and complementary way of interpreting proteomic experiments.

Highlights

- Automated analysis of protein complexes in proteomic experiments.
- Quantitative measurement of the coordinated changes in protein complex components.
- Interactive visualizations for exploratory analysis of proteomic results.

ComplexBrowser: A Tool for Identification and Quantification of Protein Complexes in Large-scale Proteomics Datasets^{*}

Wojciech Michalak[§], Vasileios Tsiamis,  Veit Schwämmle[¶],
and  Adelina Rogowska-Wrzesińska[¶]

We have developed ComplexBrowser, an open source, online platform for supervised analysis of quantitative proteomic data (label free and isobaric mass tag based) that focuses on protein complexes. The software uses manually curated information from CORUM and Complex Portal databases to identify protein complex components. For the first time, we provide a Complex Fold Change (CFC) factor that identifies up- and downregulated complexes based on the level of complex subunits coregulation. The software provides interactive visualization of protein complexes' composition and expression for exploratory analysis and incorporates a quality control step that includes normalization and statistical analysis based on the limma package. ComplexBrowser was tested on two published studies identifying changes in protein expression within either human adenocarcinoma tissue or activated mouse T-cells. The analysis revealed 1519 and 332 protein complexes, of which 233 and 41 were found coordinately regulated in the respective studies. The adopted approach provided evidence for a shift to glucose-based metabolism and high proliferation in adenocarcinoma tissues, and the identification of chromatin remodeling complexes involved in mouse T-cell activation. The results correlate with the original interpretation of the experiments and provide novel biological details about the protein complexes affected. ComplexBrowser is, to our knowledge, the first tool to automate quantitative protein complex analysis for high-throughput studies, providing insights into protein complex regulation within minutes of analysis. *Molecular & Cellular Proteomics* 18: 2324–2334, 2019. DOI: 10.1074/mcp.TIR119.001434.

Proteomics has become one of the methods of choice for large scale analysis of biological systems. Recent advances in multidimensional separation methods, improved instrument speed, sensitivity, and resolving power allow for the generation of near complete proteomes in the scope of 35 h of analysis, providing quantitative information for more than 12,000 gene products and covering 4–6 orders of magnitude

(1, 2). Proteomics provides countless research opportunities, but also challenges, especially in the domain of biological interpretation of the results. Currently, this process remains time-consuming and requires considerable expertise.

Commonly used analysis pipelines involve annotating proteins according to their molecular function, cellular component, and biological processes, based on information gathered in Gene Ontology (GO)¹ databases (3, 4). Further GO-term enrichment methods define over-represented annotations in users' data, providing a general understanding of the biological processes affected (5).

Pathway analysis is a different approach concentrated on protein specific biochemical activity. Tools such as IPA®, KEGG or Reactome map proteins to molecular pathways (6–8) and visualize processes those gene products are known to be involved in. A clear advantage of this approach is that pathway databases are mostly based on experimental, manually curated data, whereas most GO-annotations come from *in silico* predictions and text mining (9).

An alternative practice, often employed in studies of less researched organisms, is protein domain and motif analysis (10). This strategy uses sequence alignment and secondary structure prediction tools to find similarities between a protein of interest and better-annotated analogs in other species. Identification of a specific sequence motif enables the assignment of function to the previously undescribed protein.

Analysis of protein-protein interactions (PPI) is a complementary approach used very often in parallel with the above methods. Platforms such as STRING (11) use information from co-expression studies, cross-species predictions, experimental evidence, and literature mining to build protein interaction graphs, in which nodes represent gene products and edges correspond to interactions. These maps facilitate the identification of genes involved in similar processes or influenced by common regulators. Platforms for PPI investigation provide comprehensive information on proteins' function and involvement in various biological processes. How-

From the Department of Biochemistry & Molecular Biology and VILLUM Center for Bioanalytical Sciences, University of Southern Denmark, Campusvej 55, DK-5230, Odense M, Denmark

Received March 7, 2019, and in revised form, July 27, 2019

Published, MCP Papers in Press, August 25, 2019, DOI 10.1074/mcp.TIR119.001434

ever, the enormous knowledgebase of these platforms and the amount of interactions drawn from large gene/protein lists is often very difficult to handle and interpret.

Protein complexes are molecular machines that perform many of the key biochemical activities essential to the cell e.g. replication, transcription, translation, cell signaling, cell-cycle regulation, and oxidative phosphorylation. Their role in maintaining cell homeostasis and involvement in disease development (12) prove that the detailed characterization of protein complex expression would be very helpful in understanding the often highly intertwined processes in the cell.

Interrogation of the expression of components of known protein complexes in large scale proteomic data has been performed in several studies (13–15). It is now clear that many known protein complexes are translationally and post-translationally regulated, and therefore exhibit co-expression when compared across cell types and tissues. However, so far, no automated and user-friendly approach for the analysis of complex behavior in new datasets has been developed.

In this manuscript we present ComplexBrowser, which enables automated and quantitative analysis of protein complexes in proteomic experiments. The software interrogates CORUM (16) and EBI Complex Portal (17) databases to find known protein complexes present in a given protein list, and uses quantitative proteomic data (label free or isobaric mass tag based) and factor analysis to summarize the overall expression trends for each complex across the studied biological conditions. The re-analysis of two, previously published, large scale proteomic datasets show the large potential of the approach to gain in-depth knowledge about the regulation of protein complexes in different biological contexts.

EXPERIMENTAL PROCEDURES

Protein Complex Databases—The presented software relies on information from two established, manually curated protein complex databases: CORUM (16) and EBI Complex Portal (17) with 2693 and 2454 entries respectively, together covering 22 species (state for 24.05.2018). It is also possible to upload a user defined protein complex database; an example is provided in software manual ([supplementary File S1](#)).

Software Design

Data Input—ComplexBrowser accepts data Tables in .csv or .txt format as input. The file must contain single, unique UniProt (18) accessions in the first column and quantitative information in subsequent columns. Label free quantitative data in form of e.g. LFQ intensities (19) or isobaric tag based summarized reporter ion inten-

sities for each analyzed sample from TMT or iTRAQ (20) experiments can be used. Optionally, columns with confidence scores from statistical test results can be appended. These values must be calculated in relation to the first condition appearing in the input file. If they are not included by the user, differential expression analysis using the limma package (21) is conducted, along with FDR estimation using the qvalue R package (22). Examples of data input files are provided in [supplemental Tables S1 and S2](#). Additionally, the software provides an option for uploading the T-cell dataset as a test dataset (please see the detailed description of the source of the data in the section Test Datasets).

Quality Control—Prior to the analysis of protein complexes, the software creates data visualizations for quality control (QC) evaluation purposes. The QC visualizations include:

- Boxplot of log-transformed intensities to control for inconsistency in, for example, injection amounts.
- Missing value bar plots generated by summing up the number of missing values in each quantitative column to compare protein coverage between samples.
- Pairwise Scatter plots of log-transformed intensities of all proteins quantified in two selected conditions, displaying sample to sample correlations (Pearson, Kendall, Spearman), to test similarity between the samples.
- Histograms of coefficient of variation (CV) of protein absolute intensity measurements within each experimental condition to assess replicate variation.
- Q-value charts counting the number of differentially expressed features in relation to a set threshold.
- Volcano plots to depict the relationship between fold-changes and confidence for differentially regulated features.
- PCA results for visual comparison of all samples.

The software also implements four common, previously described normalization methods (23).

Identification and Visualization of Complex Components—To perform complex analysis additional parameters have to be set: (1) q value threshold, which will set the minimum q value considered for visualization purposes, (2) fold change threshold and (3) noise threshold used for the visualization of fold changes in complex components and for the generation of a summary report; (4) selection of database (CORUM or EBI Complex Portal); and (5) selection of species the samples were generated from. Upon pressing “Run analysis,” ComplexBrowser searches for relevant Uniprot accessions in users’ data and displays all protein complexes with at least one subunit found in the input. For complexes with at least 3 quantified subunits a complex fold change (CFC) is calculated (please see below). The composition of complexes and changes in the expression of their subunits are visualized using star graphs. Nodes (circles) denote subunits of the complex, the size of the nodes represents the degree of fold change, the colors indicate the type of regulation (red - downregulated, green - upregulated, blue - not changing, gray - not identified), thick edges (lines) indicate differentially regulated proteins. Selected components of the identified complex or the entire complex can be submitted for analysis in CoExpresso software (15), which assesses the co-regulatory behavior of arbitrary protein groups in more than 100 human cell types. CoExpresso is only compatible with human proteins, therefore the redirection to CoExpresso from ComplexBrowser is enabled only for human complexes.

Complex Expression Analysis—ComplexBrowser employs the fast-FARMS algorithm to summarize the abundance of protein complexes to determine complex fold changes (CFC). Fast-FARMS is a modified version of FARMS (24), a Bayesian factor analysis method with an assumption of Gaussian measurement noise originally implemented for Affymetrix probe level data summarization. It has previously been used to estimate protein abundance based on peptide concentrations

¹ The abbreviations used are: GO, Gene Ontology; CFC, complex fold change; CV, coefficient of variation; FARMS, Factor Analysis for Robust Microarray Summarization; FC, fold change; FDR, false discovery rate; KEGG, Kyoto Encyclopedia of Genes and Genomes; LFQ, label-free quantitation; LTQ, linear trap quadrupole; ODR, orthogonal distance regression; PCA, principal component analysis; PPI, protein-protein interaction; STRING, Search Tool for the Retrieval of Interacting Genes/Proteins; TMT, tandem mass tag; QC, quality control.

in the protein summarization process. It has proven useful in detecting outliers in peptide expression profiles and limiting their influence on protein quantitation (25). In ComplexBrowser, we created our own implementation of the fast-FARMS algorithm for performing weighted average summarization of log-transformed expression changes of protein complex subunits. Fast-FARMS in ComplexBrowser assumes that the complex subunit abundances are proportional to the protein complex concentration. Therefore, the subunit abundances can be modeled linearly with an additional Gaussian noise as: $x = \lambda z + \varepsilon$, where $x, \lambda \in R^1$, x and z are the subunit abundances and the true complex concentration in log scale respectively. λ expresses the contribution of each subunit to z and ε describes the noise. Fast-FARMS solves the above model by a maximum *a posteriori* estimation of λ that describes the covariation of x the best under minimal noise ε . The complex expression is calculated in two steps. First, log scaled intensities of all subunits of the same complex are processed in fast-FARMS, which assigns individual weights based on λ to each subunit and estimates the noise ε . Complex expression is then calculated by a weighted average summarization per condition process applied to the subunit intensities. The relative change in complex expression between two given conditions is defined as the complex fold change (CFC). ComplexBrowser also provides a summary measure, describing the amount of variability in the expression profile of a given complex. This is presented as a signal-to-noise ratio (S/N), or in short - Noise, by comparing λ with noise ε . Noise of 0 indicates perfect co-expression, whereas Noise = 1 indicates very poor correlation. The default noise threshold set in the software is 0.5.

Linearity of Complex Components Co-expression—Building on the idea of using the linearity of subunit co-expression as a measure of data quality (26), ComplexBrowser draws supplementary visualizations to investigate co-regulation between different conditions. For a selected protein complex, it takes log-transformed abundances of all its subunits in two conditions and displays them on a scatter plot, where each point corresponds to one protein. Orthogonal distance regression (ODR) is employed to determine the quality of co-expression similarity, because unlike ordinary least squares regression, ODR considers variability in both x and y values, therefore it fits a model that minimizes errors of both measurements (27). The procedure returns a single R^2 value per complex for each pair of conditions as a measure of co-expression.

Heatmaps and Hierarchical Clustering—Combined with dendrograms produced by hierarchical clustering algorithms, heatmaps allow the detection of proteins not following the common trend of a protein complex. These could be subunits that participate in several different complexes or interact with a given complex transiently. ComplexBrowser displays two different heatmaps—expression and correlation (between protein expressions). The expression heatmap displays log-transformed, mean-normalized expression values for all subunits within a selected complex across all experimental conditions. The same data input is used to calculate pairwise correlations between expression profiles of all complex subunits. Based on this information a correlation matrix is computed and displayed as the correlation heatmap. Both graphs use aggregative hierarchical clustering implemented in R's *hclust* function to provide dendrograms. The function uses distance measures and linkage functions selected by the user in the graphical interface.

Test Datasets—ComplexBrowser's performance was tested using protein quantification data from two previously published studies (28, 29). Further on in the manuscript we refer to the studies as *adenocarcinoma dataset* (29) and *T-cell dataset* (28).

The adenocarcinoma dataset investigates protein expression differences between formalin-fixed, paraffin embedded tissue samples from patients with colon cancer compared with healthy colon mucosa and nodal metastatic tumors using label-free quantitation based on

LFQ intensities. The MS proteomic data of the adenocarcinoma dataset were obtained from supplementary tables of the original publication available from the publisher's site (29). For this analysis we have discarded samples denoted as "CA2" and "NO2" to ensure an equal number of replicates in each condition. We have filtered the protein intensities table to retain proteins with at least 4 valid quantitative values within each condition. We have removed isoform identifiers from the original accession numbers and rows with non-unique identifiers were removed. This resulted in a dataset containing LFQ values for 6824 proteins from 3 conditions with 7 biological replicates each. The input file used in this study can be found in [supplemental Table S1](#).

The T-cell dataset studies activation of quiescent mouse T cells over four time points (0, 2, 8, and 16 h) in two biological replicates. Proteins were quantified using tandem mass tag (TMT) labeling and were analyzed on an Orbitrap Elite MS instrument. The data were obtained from the original publication from the PRIDE database (accession numbers PXD004367 and PXD005492) (30). The dataset contained normalized intensities for 8,431 proteins. The input file used in this study can be found in [supplemental Table S2](#).

Software Implementation—ComplexBrowser was implemented in R (31). The user interface was developed using Shiny, Plotly, networkD3, heatmaply, DT and data.table libraries, allowing interactive and adjustable data visualization. PreprocessCore, stringr, prisma, dplyr, limma, and qvalue packages were used for data manipulation and statistical analysis.

Software Accessibility—The tool can be accessed via web service at <http://computproteomics.bmb.sdu.dk/Apps/ComplexBrowser> or can be run locally after installation of Rstudio and the required libraries.

A fully functional demo version of ComplexBrowser is available online via <http://computproteomics.bmb.sdu.dk/Apps/Complex-Browser/>.

The source code can be downloaded from: <https://bitbucket.org/michalakw/complexbrowser>.

RESULTS

We developed ComplexBrowser to enable the identification and quantification of protein complexes in large scale proteomic experiments. The unique feature of the software is its ability to quantify changes in the abundance of protein complexes and the co-expression of their components in different conditions of the experiment. The general analysis pipeline implemented in the program is presented in Fig. 1. In brief, a table containing the quantitative information of the identified proteins is uploaded using the web browser interface. After defining the parameters of the analysis (e.g. number of conditions and replicates), the analysis of the quality of the data is carried out and visualized. In a following window the analysis of the presence and changes in abundance of protein complexes are carried out. Interactive tables and graphics allow the user to conveniently evaluate the results of the analysis. Tables containing results and a summary report are available for download. All figures generated by ComplexBrowser, as well as the QC report can be exported as vectorised graphics and edited in any PDF editor or graphics design software. An extensive description of ComplexBrowser's procedures and example results can be found in software manual in [supplementary File S1](#).

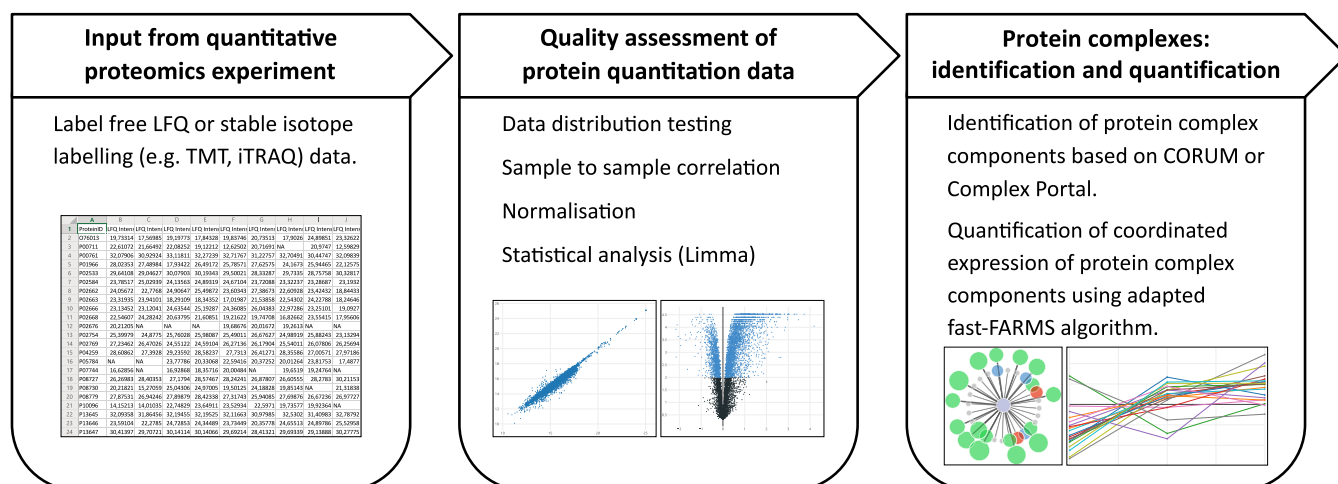


FIG. 1. **ComplexBrowser analysis workflow.** The list of identified proteins along with quantitative information is uploaded to the software and analysis parameters are set (left panel). This is followed by the assessment of the quality of the quantitative data provided (middle panel). Finally presence and changes in abundance of protein complexes are interrogated and visualized (right panel).

To test the performance of the developed platform, we have used two published proteomics studies: adenocarcinoma dataset (29) and T-cell dataset (28). The results of both the quality control and protein complex analysis steps are presented below.

Quality Control Of Proteomic Data in ComplexBrowser— The Adenocarcinoma dataset containing quantitative proteomic values from 3 biological conditions with 7 replicates each was uploaded to ComplexBrowser in the following order: C1 - control, C2 - metastasis, C3 - cancer. A QC report file generated by ComplexBrowser summarizing the quality analysis is provided in [supplementary File S2](#).

Analysis of Boxplot graphs of log-transformed intensities, Fig. 2A, indicated that normalization was necessary to reduce the variability between intensity distributions and ensure sample to sample comparability; therefore, a quantile normalization was performed. Despite the normalization, the mean CV values were 65, 78, and 77% for normal, metastasis and cancer samples respectively, Fig. 2B, indicating relatively large variability within measurements, most likely because of the clinical character of the samples.

Label free experiments may contain proteins that have been quantified in only a few samples and in other samples their quantitative value is missing (referred to as “missing values”). Many missing values in a few samples could indicate lack of technical reproducibility and/or quality of the obtained data. In the *adenocarcinoma dataset* the number of missing values per sample varied from 113 to 510 and consisted, in total, of only 3.5% of all valid measurements, Fig. 2C. It did not show any persistent bias of the data.

Differentially expressed features were determined using a paired test and FDR estimation included in the limma package, [supplemental Table S3](#). Considering the large deviation in measurements and clinical sample character, features with 0.01 FDR value were considered, resulting in the identification

of 802 proteins differentially regulated for metastasis and 813 for cancer samples. PCA analysis has shown a good separation of control samples but an overlap between the carcinoma and metastatic tissues, Fig. 2D.

Subsequently, ComplexBrowser data quality analysis was applied to the *T-cell dataset*, which consisted of four sets (0, 2, 8, and 16 h) with two replicates and was generated using TMT-based quantitation. The complete report can be found in [supplementary File S3](#). The Boxplot graph distributions of the T-cell dataset results show only very small variation in the median values for the different samples and did not require any further normalization. No missing values were observed. The variability of measurements was noticeably lower compared with the Adenocarcinoma experiment with an average CV from all conditions of 4.56% (*versus* 73.51% for the previous dataset), [supplementary File S2 and S3](#). Fig. 3A illustrates a high level of sample to sample correlation of TMT intensities for each protein between selected samples. An increasing number of differentially expressed proteins (39, 1869, and 5600) were detected after 2, 4, and 16 h of T-cell activation, at FDR of 0.05. Fig. 3B presents volcano plots generated by the ComplexBrowser software. Results of the statistical analysis downloaded from ComplexBrowser are in [supplemental Table S4](#).

Protein Complex Analysis—Complexbrowser facilitates the analysis of protein complex expression in large-scale studies. The software queries the input data for proteins reported to be complex members, investigates their co-expression patterns and visualizes results according to specified statistical, noise and expression change thresholds. It quantifies changes of complex abundance by calculating the complex fold change factor (CFC). Resulting graphs and tables allow for data exploration using interactive figures.

We recommend that setting parameters for complex analysis should be performed based on the results obtained from



FIG. 2. Examples of data quality analysis visualization of the Adenocarcinoma dataset using ComplexBrowser. A, Box plot of log-transformed LFQ intensities of all identified proteins in each sample pre- (left panel) and post- (right panel) normalization; B, CV distribution of LFQ intensities of all identified proteins within each analyzed condition; C, Column graph representing the number of missing values observed in each sample; D, Principal component analysis based on all identified and quantified proteins. Component 1 and 2 with [%] of variance explained; C1 - normal, C2 - cancer and C3 - metastatic tissue; C1_1, C1_2 etc. depict different samples within each condition.

the quality analysis module. The adenocarcinoma dataset showed a relatively high variation between samples therefore it was analyzed using the $CFC \geq 1.5$, complex Noise ≥ 0.5 and protein expression q value ≤ 0.05 . The T-cell dataset showed lower measurement variability and good correlation between the biological replicates therefore the CFC was set to ≥ 1.2 , complex noise threshold was kept at ≥ 0.5 and protein expression q value ≤ 0.05 .

The main results of the complex analysis module are presented in the form of a tabular output that can be downloaded directly from ComplexBrowser. Such output for the *adenocarcinoma* dataset for CORUM and EBI databases can be found in [supplemental Table S5](#) and [S6](#) respectively. In ComplexBrowser the table can be sorted, filtered and searched to

provide easy access to the relevant complexes. It contains complex ID, complex names, number of proteins (subunits) of the complex identified and quantified in the analyzed dataset, number of all subunits and a % complex coverage that allows the user to identify complexes that are highly represented in the analyzed dataset. CFC - complex fold change, together with Noise factor, is given for the analyzed conditions providing a means to evaluate the coordinated changes in expression of complex components. Accession numbers of all identified complex subunits and gene ontology annotations for the complex are also listed in the table.

Protein Complexes Identified in the Adenocarcinoma Dataset Reflect Biological Features of Cancer and Metastatic Tissues—Protein complex analysis of the *adenocarcinoma* data

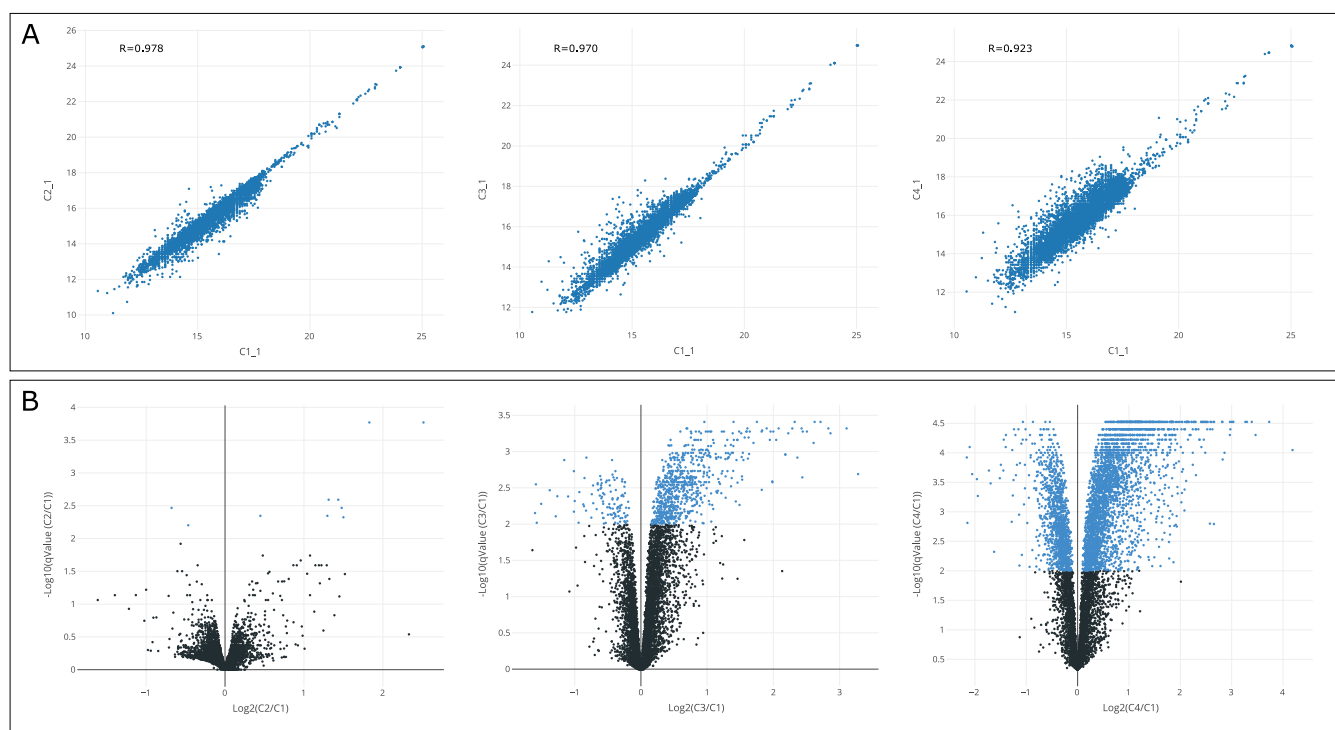


FIG. 3. Example of ComplexBrowser generated visualization of quality control analysis of the T-cell dataset. A, Pearson correlation of TMT intensities between 4 selected samples. Left panel - C1_1 versus C2_1; Middle panel - C1_1 versus C3_1; Right panel - C1_1 versus C4_1; B, Volcano plots at FDR of 0.05; C1, C2, C3 and C4 depict nonstimulated T-cells (0 h) and T-cells stimulated for 2 h, 8 h, and 16 h respectively; C1_1 and C1_2 etc. depict different replicates within each condition.

TABLE I

Summary of ComplexBrowser based analysis of protein complexes in Adenocarcinoma (CFC ≥ 1.5 , complex Noise ≥ 0.5 , protein expression q value ≤ 0.05) and T-cell datasets (CFC was set to ≥ 1.2 , complex Noise ≥ 0.5 , protein expression q value ≤ 0.05)

	Adenocarcinoma		T-cell		
Species	Homo sapiens		Mus musculus		
Quantitation method	Label-free		TMT		
Mean CV	73.51%		4.56%		
Total number of missing values (%)	3.55%		0%		
Number of conditions	3		4		
Number of replicates	7		2		
Number of proteins in the dataset	6824		8431		
Number of complexes (No. of proteins participating in complexes)					
CORUM	1519 (1687)		332 (565)		
Complex Portal	366 (437)		364 (553)		
Regulated in CORUM					
Condition	Cancer/Normal	Metastasis/Normal	2/0 h	8/0 h	16/0 h
Up	228	219	0	10	33
Down	8	9	1	3	9
Regulated in EBI Complex Portal					
Up	59	52	1	6	29
Dow	11	5	0	1	5

set identified 1519 protein complexes from CORUM and 366 from Complex Portal, Table I. Typical visualization of a protein complex and its components generated by ComplexBrowser is presented in Fig. 4. The top 5 most up-regulated and top 5 most downregulated complexes in cancer tissue selected based on the CFC are shown in Table II. Mitochondrial respi-

ratory chain I (-4.368 CFC) and F1F0- cytochrome C oxidase (-4.188 CFC) were identified as the key complexes down-regulated in both metastatic and cancer samples, Table II. This finding combined with a significant down-regulation of ATP synthase (-1.263 CFC) indicates a shift in cancer cells' metabolism from oxidative phosphorylation to glycolytic path-

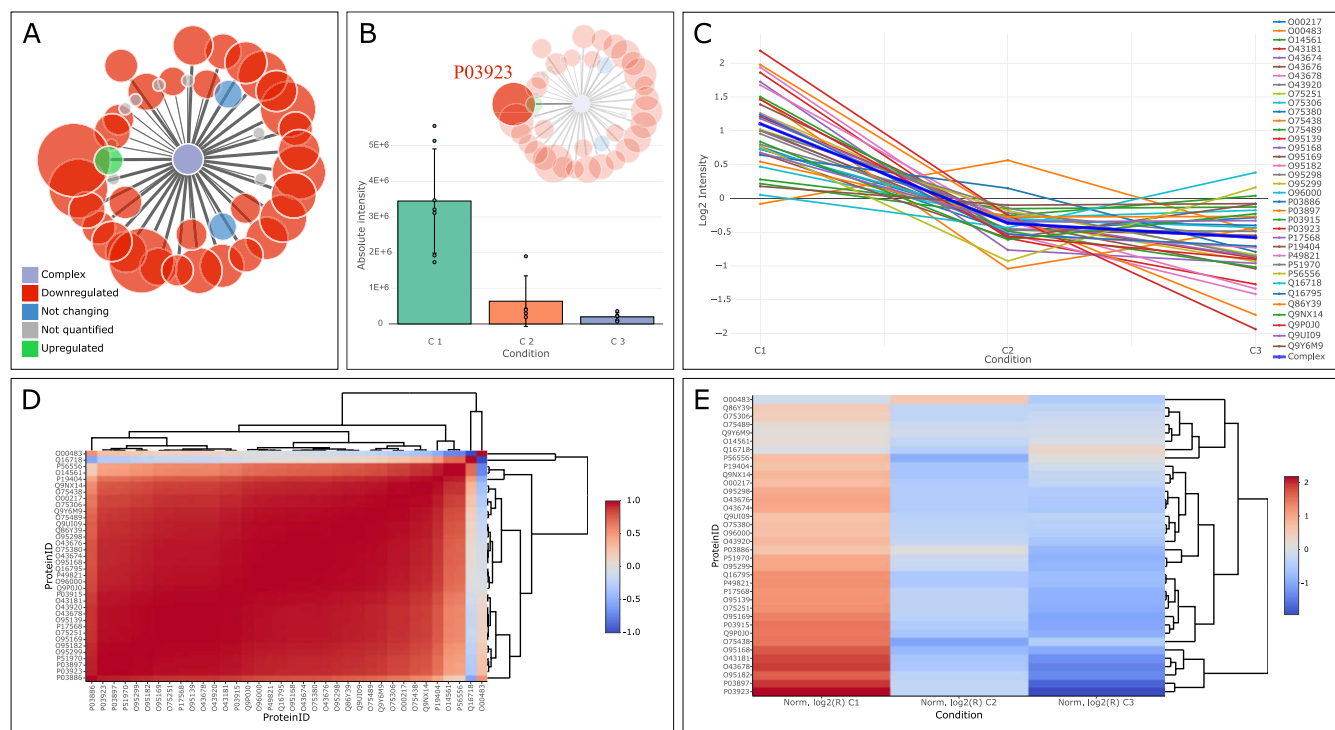


FIG. 4. Visualization of Respiratory Chain Complex I (holoenzyme) identified in Adenocarcinoma dataset using ComplexBrowser; A, Star graph representing complex components and their changes between metastasis and control samples; nodes (circles) denote subunits of the complex; the size of the nodes represents the degree of fold change; the colors indicate the type of regulation (red - downregulated, green - up-regulated, blue - not changing, gray - not identified); thick edges (lines) indicate differentially regulated proteins; B, Expression of NADH-ubiquinone oxidoreductase chain 6 (SwissProt: P03923) a subunit of Respiratory Chain Complex I showing decreased expression in cancer and metastatic tissues; C, Expression profiles of all identified and quantified subunits of the Respiratory Chain Complex I presented in panel A, thick blue line corresponds to complex expression; D, Pearson correlation heat map visualizing log-transformed, mean normalized intensities of all identified and quantified subunits of the Respiratory Chain Complex I presented in panel A in the three analyzed conditions. E, Heat map visualizing log-transformed, mean normalized intensities of all identified and quantified subunits of the Respiratory Chain Complex I presented in panel A in the three analyzed conditions; C1 - normal, C2 - cancer and C3 - metastatic tissue.

TABLE II

Top 5 up and downregulated protein complexes in Adenocarcinoma dataset based on CORUM database; NQS - number of quantified subunits; NUS - number of unique subunits; CFC Met/Cont. - complex fold change between metastatic and control samples; CFC Cancer/Cont. - complex fold change between cancer and control samples

Complex name	NQS/NUS	CFC Met/Cont.	CFC Cancer/Cont.
RaiBP1-CDC2-CCNB1 complex	3/3	6.249	7.420
CDC2-CCNA2-CDK2 complex	3/3	5.759	6.031
PeBoW complex	3/3	5.081	5.659
DDX27-PeBoW complex	4/4	5.017	5.614
Cell cycle kinase complex CDC2	3/6	4.453	4.820
GPR56-CD81-Galpha(q/11)-Gbeta complex	4/5	-1.686	-1.832
ITGA5-ITGB3-COL6A3 complex	3/3	-1.522	-2.788
Respiratory chain complex I (early intermediate NDUF4F1 assembly), mitochondrial	7/7	-4.140	-3.569
Cytochrome c oxidase, mitochondrial	13/14	-3.020	-4.188
Respiratory chain complex I (holoenzyme), mitochondrial	36/44	-3.688	-4.368

ways, which is consistent with the results of the original publication (29).

Among the top 5 regulated complexes we have identified over-expression of 3 complexes related to mitotic cell cycle progression and activation: RaiBP1-CDC2-CCNB1 complex (7.420 CFC), CDC2-CCNA2-CDK2 complex (6.031 CFC) and Cell cycle kinase

complex CDC2 (4.820 CFC), for more details see Table II and supplemental Table S4. This points to increased activation of cell division and replication, as well as reduction in respiration, which are known characteristics of cancer (32, 33).

In addition to previously described results, ComplexBrowser detected a 3.58-fold up-regulation of MTA1 complex

TABLE III

Top 10 up and downregulated protein complexes in the T-cell experiment 16 h after activation based on CORUM and EBI Complex Portal databases, result for 2 h and 8 h after activation are shown for comparison; NQS - number of quantified subunits; NUS - number of unique subunits; CFC 2/0 h, CFC 8/0 h and CFC 16/0 h—complex fold change between 2, 8, and 16 h after activation and control samples (0 h) respectively

Complex name	NQS/NUS	CFC 2/0 h	CFC 8/0 h	CFC 16/0 h
CORUM				
p19-Cdk4-cyclinD2 complex	3/3	1.074	3.189	3.674
Parvulin-associated pre-rRNP complex	49/62	−1.032	1.440	2.810
PCNA-DNA ligase complex	4/4	1.130	1.436	2.197
9S-cytosolic aryl hydrocarbon (Ah) receptor non-ligand activated complex	3/4	1.047	1.377	1.917
Bcl-xL-p53-PUMA complex, DNA damage induced	3/3	1.052	1.431	1.862
G protein complex (Btk, Gng2, Gnb1)	3/3	1.009	−1.097	−1.284
Stx7-Unc13d-Vamp8 complex	3/3	−1.086	−1.144	−1.33
Itgav-Itgb3-Gsn complex	3/3	1.028	−1.08	−1.361
Cd3d-Cd3g-Cd3e-Cd247 complex	4/4	−1.065	−1.282	−1.385
Cd3g-Cd3e-Cd247-Canx complex	4/4	−1.097	−1.280	−1.414
EBI Complex Portal				
epsilon DNA polymerase complex	4/3	1.024	1.396	2.097
Ribonucleoside-diphosphate reductase RR1 complex, RRM2 variant	8/4	1.085	−1.03	1.905
B-WICH chromatin remodeling complex	9/7	−1.104	1.221	1.837
mCRD-poly(A)-bridging complex	5/5	−1.104	1.305	1.77
AP-1 transcription factor complex FOS-JUN-NFATC2	3/3	1.94	2.25	1.701
Shelterin complex	8/8	−1.067	−1.111	−1.16
RXRalpha-RARalpha-NCOA2 retinoic acid receptor complex	4/3	1.027	−1.033	−1.175
SMAD3-SMAD4 complex	3/3	−1.049	−1.139	−1.275
SMARCA3 - Annexin A2 - S100-A10 complex	6/6	−1.016	−1.162	−1.275
AHNAK - Annexin A2 - S100-A10 complex	7/5	−1.067	−1.173	−1.329

involved in metastatic tumor formation in nodal tissue samples (34). The same complex was changed 2.33-fold in the main tumor tissue, [supplemental Table S4](#).

Activation of Mouse T-cells is Reflected in Coordinated Changes of Protein Complex Components—ComplexBrowser was used to analyze protein complexes during activation of murine T-cells. The dataset consists of 4 consecutive time points (0, 2, 8, and 16 h) collected after activation. It is expected that trends of coordinated changes of protein complexes would follow the timeline of the activation events and would be reflected in the CFC changes.

Despite the large number (8431) of proteins present in the T-cell dataset, complex analysis had identified only 332 and 374 protein complexes in the CORUM and EBI Complex Portal respectively, Table II. This is most likely because there are fewer mouse specific protein complexes present in those databases. Additionally, the complexes identified with the two databases are very different and share only 16% of the involved proteins.

Despite those drawbacks, use of both databases identified an increasing number of significantly changing complexes over the different time points of T-cell activation reflecting the gradual changes in the proteome of activated cells. For example, based on EBI Complex Portal, one complex was significantly regulated 2 h after T-cell activation and 7 and 34 at 8 and 16 h time points respectively, Table II. A summary of the

top 5 most up-regulated and top 5 most downregulated complexes after 16 h of T-cell activation can be found in Table III.

The largest quantified complex was Parvulin-associated pre-rRNP complex (CORUM) with 49 out of 62 subunits quantified, Fig. 5A. Its abundance was highly correlated (Noise = 0.001) and increased over the time course of the experiment (−1.03, 1.44, 2.81 CFC after 2, 8, and 16 h of activation respectively). This protein complex is involved in ribosome biogenesis and contains several ribosomal subunits (35). BAT3 complex (EBI Complex Portal), responsible for the targeting of the transmembrane domain-containing proteins from the ribosome toward membranes, was also up-regulated in the last condition (CFC 1.42), Fig. 5B, suggesting an overall increase in processes related to protein synthesis.

Combined data from CORUM and EBI Complex Portal showed that 62 complexes were induced 16 h after activation, with a substantial increase in abundance between 8 h and 16 h after activation, [supplemental Table S7 and S8](#).

The p19-Cdk4-cyclinD2 (3.96 CFC after 16 h) and Cyclin D1-associated (2.47 CFC after 16 h) protein complexes were found to be among the most up-regulated 8 and 16 h after activation. These two complexes are involved in the regulation of cell cycle and transition through G1 phase, and their coordinated up-regulation reflects the transition of T-cells into the proliferative state. These changes were accompanied by an increase in the expression of various DNA polymerase com-

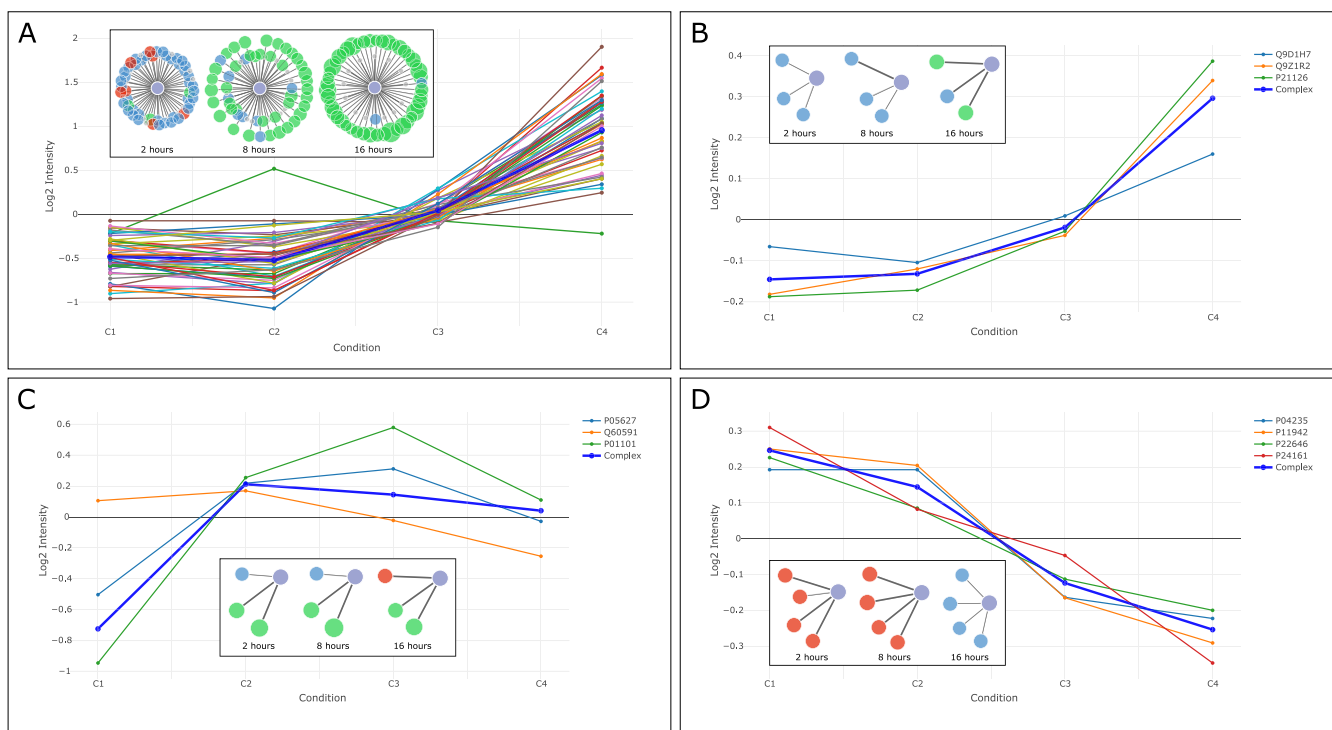


FIG. 5. Visualization of selected protein complexes identified in T-cell dataset using ComplexBrowser. Inserts contain star graphs representing complex components and their changes during T-cell differentiation; nodes (circles) denote subunits of the complex; the size of the nodes represents the degree of fold change; the colors indicate the type of regulation (red - downregulated, green - up-regulated, blue - not changing, gray - not identified); thick edges (lines) indicate differentially regulated proteins. Graphs are normalized expression profiles visualizing the expression of the subunits of complexes at 2, 8 and 16 h after T-cell activation, whereas complex expression is illustrated with a thick blue line. C1, C2, C3 and C4 depict non-stimulated T-cells (0 h) and T-cells stimulated for 2 h, 8 h and 16 h respectively. A, Parvulin-associated pre-rRNP complex; B, BAT3 complex; C, AP-1 transcription factor complex FOS-JUN-NFATC2; D - Cd3d-Cd3g-Cd3e-Cd247 complex.

plexes e.g. DNA synthesome (1.49 CFC after 16 h), DNA polymerase alpha, delta and epsilon complexes (1.45, 1.25, 2.09 CFC after 16 h), and Telomerase holoenzyme complex (1.40 CFC after 16 h) pointing to an increase in processes associated with DNA replication and cell proliferation.

The AP-1 transcription factor complex FOS-JUN-NFATC2 (CORUM) (CFC 1.87 at 2 h, 2.13 at 8 h and 1.62 at 16 h) is an example of a complex with a highly coordinated early response to the stimulus, Fig. 5C. Similar patterns in the AP-1 complex were also reported by Tan *et al.* (28).

A trend of downregulation was found in the expression of the Cd3d-Cd3g-Cd3e-Cd247 complex (1.07, -1.2, -1.36 CFC), a membrane glycoprotein assembly which is a part of the T-cell co-receptor, Fig. 5D, that is shut down during T-cell activation (28).

Additionally, we have identified a variety of complexes involved in post-translational modification of histones and chromatin remodeling, the majority of which showed increased expression at 8 and 16 h of T-cell activation, Table IV. These complexes and regulation of epigenetic processes in the maturation of T-cells had been largely overlooked in the original study. Describing the regulation of transcription by post-translational modifications and chromatin remodeling requires

their quantification which was not done in this case. An extensive follow-up analysis is necessary to interpret these results.

In the *T-cell* dataset ComplexBrowser analysis identified upregulated complexes involved in DNA replication and chromatin remodeling, protein synthesis initiation and cell cycle progression, which allowed us to conclude that the T-cells undergo significant cellular reprogramming and exit a quiescent state 8 to 16 h after stimulation. Moreover, there was a decrease in T-cell signaling receptor expression from the first time point of the experiment.

DISCUSSION

ComplexBrowser is, to our knowledge, the first automated tool that enables quantitative analysis of protein complexes in proteomic experiments. It is available through a web-browser and does not require any installations or programming experience. Thus, it has high potential for integration into data analysis workflows commonly used by the scientific community.

Based on the datasets analyzed in this manuscript we have shown that it is compatible with label free and TMT based quantitation experiments. And there are no technical limita-

TABLE IV

Chromatin remodelling and histone modifications associated protein complexes identified in T-cells 16 h after activation; NQS - number of quantified subunits; NUS - number of unique subunits; CFC 2/0 h, CFC 8/0 h, and CFC 16/0 h—complex fold change between 2, 8, and 16 h after activation and control samples (0 h) respectively

Complex name	NQS/NUS	CFC 2/0 h	CFC 8/0 h	CFC 16/0 h
B-WICH chromatin remodelling complex	9/7	−1.104	1.221	1.837
Methylosome	3/3	1.016	1.185	1.689
Chromatin assembly factor 1 complex	3/3	1.03	−1.031	1.635
CHRAAC chromatin remodeling complex	4/4	−1.032	1.138	1.52
eNoSc complex	3/3	−1.075	1.046	1.371
BAT3 complex	3/3	1.009	1.091	1.36
NuA4 histone acetyltransferase complex	19/19	−1.002	1.108	1.321
PPP4C-PPP4R2-PPP4R3B protein phosphatase 4 complex	3/3	1.058	1.15	1.268
MBD3/NuRD nucleosome remodeling and deacetylase complex	12/12	−1.135	1.04	1.261
SRCAP histone exchanging complex	9/8	−1.001	1.073	1.252

tions that prevent it from being used with any other MS based protein quantitation method or even gene expression data. ComplexBrowser can handle large proteomic studies with over 8000 quantified proteins and is able to display summary results within 1 min from data input. Interactive visualizations provide an intuitive tool for exploratory analysis and data interpretation, enabling the user to investigate the behavior of whole complexes, as well as single subunits.

The CFC effectively assists in finding complexes that are changing in expression in a synchronized manner and is a measure of complex behavior. Subunits which are not coherent with trends in complex expression are also easily identified using the extensive visualization tools implemented in the software.

In both test datasets ComplexBrowser identified the key protein complexes that are known to be regulated in cancer and during T-cell activation. The biological interpretation enabled by the interrogation of protein complexes agreed with the conclusions drawn based on GO-annotations in the original studies. The tool also added new insights based on the investigation of annotated protein complexes that were previously not considered in the analysis.

The novelty of the approach presented in ComplexBrowser is that in contrast to GO annotation and GO enrichment analysis ComplexBrowser identifies components of protein complexes from manually curated databases *e.g.* CORUM, Complex Portal or user defined list of complexes. A completely new feature, not available in any other software facilitating quantitative analysis of proteomic data is the application of the fast-FARMS algorithm (36) for providing a quantitative measure of the changes in complex components, in the form of the CFC factor and evaluation of the coordinated expression of complex subunits in the form of Noise. Thus, ComplexBrowser provides a complementary approach to, for example, STRING or GO-term enrichment tools. More studies, including different biological perturbations need to be analyzed in ComplexBrowser to obtain a complete picture of its utility.

ComplexBrowser relies on information stored in CORUM (16) and Complex Portal databases (17) and is therefore dependent on the efforts of their administrators. The composition of these resources introduces a bias in the analysis, because the largest proportion of complexes described in both databases are of human origin (66.36% of CORUM and 25.79% of Complex Portal). Thus, currently, ComplexBrowser is most suitable for analysis of human proteins. This is visible when comparing the number of proteins found to be involved in complexes from the adenocarcinoma (human) and T-cell (mouse) datasets, Table I. Additionally the databases contain entries that are not fully annotated. Further developments of the databases will improve the results provided by the software.

Acknowledgments—We thank Ole N. Jensen and Lauren Elizabeth Smith for their critical comments to the project and manuscript.

DATA AVAILABILITY

The source code can be downloaded from: <https://bitbucket.org/michalakw/complexbrowser>.

* A.R.-W. was supported by a grant from the Independent Research Fund Denmark - Natural Sciences and VILLUM Foundation for a grant to the VILLUM Center for Bioanalytical Sciences at SDU. V.S. was supported by ELIXIR DK. W.M. was supported by student grant from MC2 Therapeutics ApS.

§ This article contains [supplemental Figures and Tables](#).

‡ To whom correspondence should be addressed: Department of Biochemistry and Molecular Biology and VILLUM Center for Bioanalytical Sciences, University of Southern Denmark, Campusvej 55, DK-5230, Odense M, Denmark. E-mail: adelinar@bmb.sdu.dk.

§ Current affiliation: Biostatistics NADs & Complications, Novo Nordisk A/S, Vandtårnsvej 114, DK-2860 Søborg, Denmark.

¶ These authors contributed equally to this work.

Author contributions: W.M., V.T., V.S., and A.R.-W. designed research; W.M., V.T., and V.S. performed research; W.M., V.T., V.S., and A.R.-W. analyzed data; W.M., V.T., V.S., and A.R.-W. wrote the paper.

REFERENCES

1. Bekker-Jensen, D. B., Kelstrup, C. D., Batth, T. S., Larsen, S. C., Haldrup, C., Bramsen, J. B., Sørensen, K. D., Høyer, S., Ørntoft, T. F., Andersen,

- C. L., Nielsen, M. L., and Olsen, J. V. (2017) An optimized shotgun strategy for the rapid generation of comprehensive human proteomes. *Cell Systems* **4**, 587–599.e584
2. Kelstrup, C. D., Bekker-Jensen, D. B., Arrey, T. N., Hogrebe, A., Harder, A., and Olsen, J. V. (2018) Performance evaluation of the Q Exactive HF-X for shotgun proteomics. *J. Proteome Res.* **17**, 727–738
3. Ashburner, M., Ball, C. A., Blake, J. A., Botstein, D., Butler, H., Cherry, J. M., Davis, A. P., Dolinski, K., Dwight, S. S., Eppig, J. T., Harris, M. A., Hill, D. P., Issel-Tarver, L., Kasarskis, A., Lewis, S., Matese, J. C., Richardson, J. E., Ringwald, M., Rubin, G. M., and Sherlock, G. (2000) Gene ontology: tool for the unification of biology. The Gene Ontology Consortium. *Nat. Genet.* **25**, 25–29
4. The Gene Ontology, C. (2017) Expansion of the Gene Ontology knowledgebase and resources. *Nucleic Acids Res.* **45**, D331–D338
5. Eden, E., Navon, R., Steinfeld, I., Lipson, D., and Yakhini, Z. (2009) GOrilla: a tool for discovery and visualization of enriched GO terms in ranked gene lists. *BMC Bioinformatics* **10**, 48
6. Kramer, A., Green, J., Pollard, J., Jr, and Tugendreich, S. (2014) Causal analysis approaches in Ingenuity Pathway Analysis. *Bioinformatics* **30**, 523–530
7. Ogata, H., Goto, S., Sato, K., Fujibuchi, W., Bono, H., and Kanehisa, M. (1999) KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Res.* **27**, 29–34
8. Fabregat, A., Sidiropoulos, K., Garapati, P., Gillespie, M., Hausmann, K., Haw, R., Jassal, B., Jupe, S., Korminger, F., McKay, S., Matthews, L., May, B., Milacic, M., Rothfels, K., Shamovsky, V., Webber, M., Weiser, J., Williams, M., Wu, G., Stein, L., Hermjakob, H., and D'Eustachio, P. (2016) The Reactome pathway Knowledgebase. *Nucleic Acids Res.* **44**, D481–D487
9. Rhee, S. Y., Wood, V., Dolinski, K., and Draghici, S. (2008) Use and misuse of the gene ontology annotations. *Nat. Rev. Genet.* **9**, 509–515
10. Schmidt, A., Forne, I., and Imhof, A. (2014) Bioinformatic analysis of proteomics data. *BMC Syst. Biol.* **8**, S3
11. Szklarczyk, D., Morris, J. H., Cook, H., Kuhn, M., Wyder, S., Simonovic, M., Santos, A., Doncheva, N. T., Roth, A., Bork, P., Jensen, L. J., and von Mering, C. (2017) The STRING database in 2017: quality-controlled protein-protein association networks, made broadly accessible. *Nucleic Acids Res.* **45**, D362–D368
12. David, A., Razali, R., Wass, M. N., and Sternberg, M. J. (2012) Protein-protein interaction sites are hot spots for disease-associated nonsynonymous SNPs. *Hum. Mutat.* **33**, 359–363
13. Ori, A., Iskar, M., Buczak, K., Kastiris, P., Parca, L., Andres-Pons, A., Singer, S., Bork, P., and Beck, M. (2016) Spatiotemporal variation of mammalian protein complex stoichiometries. *Genome Biol.* **17**, 47
14. Goncalves, E., Fragoulis, A., Garcia-Alonso, L., Cramer, T., Saez-Rodriguez, J., and Beltrao, P. (2017) Widespread post-transcriptional attenuation of genomic copy-number variation in cancer. *Cell Syst.* **5**, 386–398.e384
15. Chalabi, M. H., Tsiamis, V., Käll, L., Vandin, F., and Schwämmle, V. (2019) CoExpresso: assess the quantitative behavior of protein complexes in human cells. *BMC Bioinformatics* **20**, 17
16. Ruepp, A., Waagele, B., Lechner, M., Brauner, B., Dunger-Kaltenbach, I., Fobo, G., Frishman, G., Montrone, C., and Mewes, H. W. (2010) CORUM: the comprehensive resource of mammalian protein complexes—2009. *Nucleic Acids Res.* **38**, D497–D501
17. Meldal, B. H., Forner-Martinez, O., Costanzo, M. C., Dana, J., Demeter, J., Dumousseau, M., Dwight, S. S., Gaulton, A., Licata, L., Melidoni, A. N., Ricard-Blum, S., Roehert, B., Skzyrpek, M. S., Tiwari, M., Velankar, S., Wong, E. D., Hermjakob, H., and Orchard, S. (2015) The complex portal—an encyclopaedia of macromolecular complexes. *Nucleic Acids Res.* **43**, D479–D484
18. The UniProt, C. (2017) UniProt: the universal protein knowledgebase. *Nucleic Acids Res.* **45**, D158–D169
19. Cox, J., Hein, M. Y., Lubner, C. A., Paron, I., Nagaraj, N., and Mann, M. (2014) Accurate proteome-wide label-free quantification by delayed normalization and maximal peptide ratio extraction, termed MaxLFQ. *Mol. Cell Proteomics* **13**, 2513–2526
20. Rauniyar, N., and Yates, J. R., 3rd. (2014) Isobaric labeling-based relative quantification in shotgun proteomics. *J. Proteome Res.* **13**, 5293–5309
21. Ritchie, M. E., Phipson, B., Wu, D., Hu, Y., Law, C. W., Shi, W., and Smyth, G. K. (2015) limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Res.* **43**, e47
22. Storey, J. D. (2002) A direct approach to false discovery rates. *J. Roy. Stat. Soc. B* **64**, 479–498
23. Chawade, A., Alexandersson, E., and Levander, F. (2014) Normalizer: A tool for rapid evaluation of normalization methods for omics data sets. *J. Proteome Res.* **13**, 3114–3120
24. Hochreiter, S., Clevert, D. A., and Obermayer, K. (2006) A new summarization method for affymetrix probe level data. *Bioinformatics* **22**, 943–949
25. Zhang, B., Pirmoradian, M., Zubarev, R., and Kall, L. (2017) Covariation of peptide abundances accurately reflects protein concentration differences. *Mol. Cell Proteomics* **16**, 936–948
26. Rogowska-Wrzesinska, A., Wrzesinski, K., and Fey, S. J. (2014) Heteromer score-using internal standards to assess the quality of proteomic data. *Proteomics* **14**, 1042–1047
27. Boggs, P. T., Spiegelman, C. H., Donaldson, J. R., and Schnabel, R. B. (1988) A computational examination of orthogonal distance regression. *J. Econometrics* **38**, 169–201
28. Tan, H. Y., Yang, K., Li, Y. X., Shaw, T. I., Wang, Y. Y., Blanco, D. B., Wang, X. S., Cho, J. H., Wang, H., Rankin, S., Guy, C., Peng, J. M., and Chi, H. B. (2017) Integrative proteomics and phosphoproteomics profiling reveals dynamic signaling networks and bioenergetics pathways underlying T cell activation. *Immunity* **46**, 488–503
29. Wisniewski, J. R., Ostasiewicz, P., Dus, K., Zielinska, D. F., Gnad, F., and Mann, M. (2012) Extensive quantitative remodeling of the proteome between normal colon tissue and adenocarcinoma. *Mol. Syst. Biol.* **8**, 611
30. Perez-Riverol, Y., Csordas, A., Bai, J., Bernal-Llinares, M., Hewapathirana, S., Kundu, D. J., Inuganti, A., Griss, J., Mayer, G., Eisenacher, M., Perez, E., Uszkoreit, J., Pfeuffer, J., Sachsenberg, T., Yilmaz, S., Tiwary, S., Cox, J., Audain, E., Walzer, M., Jarnuczak, A. F., Ternent, T., Brazma, A., and Vizcaino, J. A. (2019) The PRIDE database and related tools and resources in 2019: improving support for quantification data. *Nucleic Acids Res.* **47**, D442–D450
31. Team, R. C. (2018) R: A language and environment for statistical computing. *R Foundation for Statistical Computing, Vienna, Austria*
32. Solaini, G., Sgarbi, G., and Baracca, A. (2011) Oxidative phosphorylation in cancer cells. *BBA-Bioenergetics* **1807**, 534–542
33. Casimiro, M. C., Crosariol, M., Loro, E., Li, Z., and Pestell, R. G. (2012) Cyclins and cell cycle control in cancer and disease. *Genes Cancer* **3**, 649–657
34. Yao, Y. L., and Yang, W. M. (2003) The metastasis-associated proteins 1 and 2 form distinct protein complexes with histone deacetylase activity. *J. Biol. Chem.* **278**, 42560–42568
35. Fujiyama, S., Yanagida, M., Hayano, T., Miura, Y., Isobe, T., Fujimori, F., Uchida, T., and Takahashi, N. (2002) Isolation and proteomic characterization of human Parvulin-associating preribosomal ribonucleoprotein complexes. *J. Biol. Chem.* **277**, 23773–23780
36. Perez-Alvarez, S., Gomez, G., and Brander, C. (2015) FARMS: A new algorithm for variable selection. *Biomed. Res. Int.* **2015**, 319797